

Singular Transformations of Probability Density Functions

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Overview

For non-singular transformations of variables with distributions, there are standard formulas that describe the probability distribution of the target variable. However, in the case of singular transformations, there is no immediate formula. Often what is done is to add independent variables in a somewhat ad hoc fashion to make the transformation non-singular, and then “integrate out” – if possible – the independent variables from the resulting formula. Below we describe a procedure to determine the target distribution with a more systematic process.

Singular Transformation Density Formula

We state and derive a theorem for constructing the probability density function of a random variable Y which is a singular function of a random variable X . By singular, we mean that $Y = f(X)$ with f a function such that $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ and $m < n$.

Theorem 1. *Let $f : \mathcal{D} \subset \mathbf{R}^n \rightarrow \mathbf{R}^m$ be a continuously differentiable function with $n > m$, such that $Df(\mathbf{x})$ has rank m for all $\mathbf{x} \in \mathcal{D}$, and let P_X be a continuous probability density function of the random variable X which vanishes outside $\mathcal{D}$. Then each nonempty level set $f^{-1}(\mathbf{y})$ is an $(n - m)$ -dimensional C^1 manifold, and the random variable $Y = f(X)$ has a density P_Y given for almost every $\mathbf{y}$ by the surface integral*

$$P_Y(\mathbf{y}) = \int_{f^{-1}(\mathbf{y})} \frac{P_X(\mathbf{x})}{\sqrt{|Df(\mathbf{x}) Df^T(\mathbf{x})|}} dS \quad (1)$$

where dS denotes $(n - m)$ -dimensional surface measure on $f^{-1}(\mathbf{y})$. If, in addition, the integral in (1) converges locally uniformly in $\mathbf{y}$, then this version of P_Y is continuous and (1) holds for every $\mathbf{y}$.

For computation we use a parameterized form of (1). If $f^{-1}(\mathbf{y})$ is covered – up to a set of surface measure zero – by $k(\mathbf{y})$ disjoint injective C^1 parameterizations $\mathbf{x}_i(\mu; \mathbf{y})$, $i \in [1, k(\mathbf{y})]$, $\mu \in \Omega_i \subset \mathbf{R}^{n-m}$, then (1) becomes

$$P_Y(\mathbf{y}) = \sum_{i=1}^{k(\mathbf{y})} \int_{\Omega_i} \frac{P_X(\mathbf{x}_i(\mu; \mathbf{y}))}{\sqrt{|Df(\mathbf{x}_i(\mu; \mathbf{y})) Df^T(\mathbf{x}_i(\mu; \mathbf{y}))|}} \sqrt{|D_\mu \mathbf{x}_i^T(\mu; \mathbf{y}) D_\mu \mathbf{x}_i(\mu; \mathbf{y})|} d\mu \quad (2)$$

Formula (1) is a form of the smooth coarea formula of geometric measure theory [1, 2]. The aim of this note is a self-contained derivation and a collection of worked examples.

Prerequisite Results

We recall how to compute the pseudo inverse of a matrix which maps from a high dimension to a low dimension.

Lemma 1. *Let $A : \mathbf{R}^n \rightarrow \mathbf{R}^m$ be a linear transformation with $n > m$. If the rank of A is m , then the pseudo inverse of A is $A^T(AA^T)^{-1}$.*

Proof. A given $\mathbf{x} \in \mathbf{R}^n$ can be written uniquely as $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2$ where $\mathbf{x}_1 \in R(A^T)$ and $\mathbf{x}_2 \in N(A)$. For a given $\mathbf{y} \in \mathbf{R}^m$, it is easy to show that there is a unique $\mathbf{x}_p \in R(A^T)$ such that $A\mathbf{x}_p = \mathbf{y}$. The set of all $\mathbf{x} \in \mathbf{R}^n$ such that $A\mathbf{x} = \mathbf{y}$ has the form $\{\mathbf{x}_p + \mathbf{x}_n \mid \mathbf{x}_n \in N(A)\}$. The pseudo inverse of A is the map that sends $\mathbf{y}$ to $\mathbf{x}_p$. Since $R(A^T) \perp N(A)$, $\mathbf{x}_p$ is the solution to the problem:

$$\begin{aligned} & \min_{\mathbf{x}} \|\mathbf{x}\|^2 \\ & \text{subject to : } \mathbf{y} = A\mathbf{x} \end{aligned}$$

The solution of this can be found using Lagrange multipliers. The resulting minimization problem is:

$$\min_{\mathbf{x}} F(\mathbf{x}) = \min_{\mathbf{x}} \|\mathbf{x}\|^2 + \langle \lambda, \mathbf{y} - A\mathbf{x} \rangle$$

At a minimum, $\mathbf{x}^*$, $DF(\mathbf{x}^*)(\mathbf{h}) = \mathbf{0}$. $DF(\mathbf{x}^*)$ can be computed from $F(\mathbf{x} + \mathbf{h}) = F(\mathbf{x}) + DF(\mathbf{x})(\mathbf{h}) + o(\mathbf{h})$. The order $\mathbf{h}$ terms are:

$$2\langle \mathbf{x}, \mathbf{h} \rangle - \langle \lambda, A\mathbf{h} \rangle$$

Therefore, $DF(\mathbf{x}^*)(\mathbf{h}) = \langle 2\mathbf{x}^* - A^T\lambda, \mathbf{h} \rangle$. Since $DF(\mathbf{x}^*) \equiv 0$, we have that $2\mathbf{x}^* - A^T\lambda = \mathbf{0}$. Applying A to this last equation and using the fact that A has rank m and $A\mathbf{x}^* = \mathbf{y}$ yields:¹ $\lambda = 2(AA^T)^{-1}\mathbf{y}$. This implies that $\mathbf{x}^* = A^T(AA^T)^{-1}\mathbf{y}$. Since $\|\mathbf{x}\|^2$ is strictly convex and the constraint set is affine, this stationary point is indeed the unique minimum.

We need the following change of volume formula:

Lemma 2. *Let $A : \mathbf{R}^n \rightarrow \mathbf{R}^m$ be a linear transformation with $n < m$, then if A has rank n it transforms the n -dimensional unit cube in $\mathbf{R}^n$ to a region in $\mathbf{R}^m$ with n -dimensional volume $\sqrt{|A^T A|}$.*

Proof. Since $R(A)$ has dimension n , there is a linear map $P : R(A) \rightarrow \mathbf{R}^n$ which is a linear isomorphism that preserves the inner product. That is, $\langle P\mathbf{x}, P\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in R(A)$, equivalently $\langle (P^T P - I)\mathbf{x}, \mathbf{y} \rangle = 0 \forall \mathbf{x}, \mathbf{y} \in R(A)$. This says $P^T P$ acts as the identity on $R(A)$; in particular, $P^T P A = A$ since each column of A lies in $R(A)$. The change in volume of the unit cube under PA is the same as under A since P preserves volumes. However, PA is a map from $\mathbf{R}^n$ to $\mathbf{R}^n$, so it changes the volume of the unit cube by $|PA|$. Since for given transformations $B, C : \mathbf{R}^n \rightarrow \mathbf{R}^n$ we have $|B^T| = |B|$ and $|BC| = |B||C|$, it follows that $|PA|^2 = |PA||PA| = |A^T P^T||PA| = |A^T P^T P A| = |A^T A|$, where the last equality uses $P^T P A = A$. Therefore, the change in volume is $\sqrt{|A^T A|}$.

Derivation of the Singular Transformation Formula

¹ Since A has rank m ($\dim(R(A)) = m$) and $R(AA^T) = R(A)$, AA^T is invertible.

We now derive Theorem 1. The argument is geometric and informal at two points: the product decomposition near a level set is only local, and the linearization step is not carried out in full detail. A fully rigorous proof may be based on the coarea formula [1, 2].

Derivation. Let ν be the induced probability measure defined by²

$$\nu(\Omega_Y) = \int_{f^{-1}(\Omega_Y)} P_X(\mathbf{x}) d\mathbf{x}$$

Given an arbitrary Borel measurable region Ω_Y and letting $\Omega_X = f^{-1}(\Omega_Y)$ we have that

$$\nu(\Omega_Y) = \int_{\Omega_X} P_X(\mathbf{x}) d\mathbf{x} \quad (3)$$

Assume that for each $\mathbf{y} \in \Omega_Y$, $f^{-1}(\mathbf{y})$ is a single connected $(n - m)$ -dimensional manifold (i.e., $k(\mathbf{y}) = 1$). Then there are local coordinates $\mathbf{x}_1, \mathbf{x}_2$, and manifolds $\Omega_{X_1}, \Omega_{X_2}$, such that Ω_X is a product of Ω_{X_1} and Ω_{X_2} with the component manifolds $\Omega_{X_1}, \Omega_{X_2}$, locally tangent to $R(Df^T(\mathbf{x}))$ and $N(Df(\mathbf{x}))$ respectively. Therefore,

$$\nu(\Omega_Y) = \int_{\Omega_X} P_X(\mathbf{x}(\mathbf{x}_1, \mathbf{x}_2)) d\mathbf{x}_1 d\mathbf{x}_2.$$

Since for a given transformation A , $R(A^T) \perp N(A)$, it follows that the product measure $d\mathbf{x}_1 d\mathbf{x}_2$ is equal to the product of the component measures. Thus, we may write (3) as

$$\nu(\Omega_Y) = \int_{\Omega_{X_1}} \int_{\Omega_{X_2}} P_X(\mathbf{x}(\mathbf{x}_1, \mathbf{x}_2)) d\mathbf{x}_1 d\mathbf{x}_2.$$

For a given $\mathbf{x}_2$, $\mathbf{y}$ maps to the manifold Ω_{X_1} in an invertible way (locally). The linear approximation to the map f is Df ; by Lemma 1 its pseudo inverse is $Df^T(Df Df^T)^{-1}$, so that locally

$$\mathbf{x}_1 \approx Df^T(\mathbf{x}) (Df(\mathbf{x}) Df^T(\mathbf{x}))^{-1} \mathbf{y}.$$

Using $\mathbf{y}$ to express $\mathbf{x}_1$ combined with Lemma 2 we have

$$\nu(\Omega_Y) = \int_{\Omega_Y} \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}} d\mathbf{x}_2 d\mathbf{y}.$$

The integral above exhibits a Lebesgue density for ν , so ν is absolutely continuous with respect to Lebesgue measure³ and there exists a measurable function $P_Y(\mathbf{y})$ such that $\nu(\Omega_Y) = \int_{\Omega_Y} P_Y(\mathbf{y}) d\mathbf{y}$. So that

$$\int_{\Omega_Y} P_Y(\mathbf{y}) d\mathbf{y} = \int_{\Omega_Y} \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}} d\mathbf{x}_2 d\mathbf{y}$$

Or,

$$\int_{\Omega_Y} \left(P_Y(\mathbf{y}) - \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}} d\mathbf{x}_2 \right) d\mathbf{y} = 0$$

Since this is true for all Borel measurable sets, Ω_Y , the two integrands agree for almost every $\mathbf{y}$:

$$P_Y(\mathbf{y}) = \int_{\Omega_{X_2}} \frac{P_X(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))}{\sqrt{|Df(\mathbf{x}(\mathbf{y}, \mathbf{x}_2)) Df^T(\mathbf{x}(\mathbf{y}, \mathbf{x}_2))|}} d\mathbf{x}_2 \quad (4)$$

² The measure extends in a natural way to all Lebesgue measurable sets.

³ The existence of the density is the Radon–Nikodym theorem. Since Lebesgue measure is regular, the identity extends from Borel sets to all Lebesgue measurable sets.

But in this case the component manifold, Ω_{X_2} , is just $f^{-1}(\mathbf{y})$. If we parameterize this as $\mathbf{x}(\mu; \mathbf{y})$, then by Lemma 2 we have

$$P_Y(\mathbf{y}) = \int_{f^{-1}(\mathbf{y})} \frac{P_X(\mathbf{x}(\mu; \mathbf{y}))}{\sqrt{|Df(\mathbf{x}(\mu; \mathbf{y}))Df^T(\mathbf{x}(\mu; \mathbf{y}))|}} \sqrt{|D_\mu \mathbf{x}^T(\mu; \mathbf{y})D_\mu \mathbf{x}(\mu; \mathbf{y})|} d\mu$$

which is (2) with $k(\mathbf{y}) = 1$.

In the general case, the level set may have several components, or may require several charts to cover it, as with $y = x_1^2 + x_2^2$ (Example 3 below), where two parameterizations together cover the circle $f^{-1}(y)$. The contributions of the charts add, giving the sum in (2).

Finally, when the integrals in (1) and (2) converge locally uniformly in $\mathbf{y}$, they define a continuous function of $\mathbf{y}$, and we may take this version as P_Y ; this gives the continuity assertion of Theorem 1.

Applications of Theorem 1

We apply Theorem 1 with various types of singular mappings.

Example 1: $y = f(\mathbf{x}) = x_1 + x_2 \quad (f : \mathbf{R}^2 \rightarrow \mathbf{R})$

The set $f^{-1}(y)$ can be parameterized as $\mathbf{x}_1(\mu; y) = (\mu, y - \mu)^T$ with $\mu \in (-\infty, \infty)$.

$$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; y) D_\mu \mathbf{x}_1(\mu; y)|} = \|(1, -1)\| = \sqrt{2}.$$

$$\text{Also, } \sqrt{|Df(\mathbf{x}_1(\mu; y)) Df^T(\mathbf{x}_1(\mu; y))|} = \|\nabla_{\mathbf{x}} f(\mathbf{x}_1(\mu; y))\| = \|(1, 1)\| = \sqrt{2}.$$

Therefore, by Theorem 1 we have:

$$P_Y(y) = \int_{-\infty}^{\infty} P_{X_1 X_2}(\mu, y - \mu) d\mu$$

Example 2: $y = f(\mathbf{x}) = x_1/x_2 \quad (f : \mathbf{R}^2 \rightarrow \mathbf{R})$

Here the natural domain is $\mathcal{D} = \{\mathbf{x} \in \mathbf{R}^2 \mid x_2 \neq 0\}$. The set $f^{-1}(y)$ can be parameterized as $\mathbf{x}_1(\mu; y) = (y\mu, \mu)^T$ with $\mu \in (-\infty, 0) \cup (0, \infty)$. We label the first and second components of $\mathbf{x}_1$ as $x_{1,1}$ and $x_{1,2}$ respectively.

$$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; y) D_\mu \mathbf{x}_1(\mu; y)|} = \|(y, 1)\| = \sqrt{1 + y^2}. \text{ And,}$$

$$\begin{aligned} \sqrt{|Df(\mathbf{x}_1(\mu; y)) Df^T(\mathbf{x}_1(\mu; y))|} &= \|\nabla_{\mathbf{x}} f(\mathbf{x}_1(\mu; y))\| \\ &= \|(1/x_{1,2}(\mu; y), -x_{1,1}(\mu; y)/x_{1,2}(\mu; y)^2)\| \\ &= \sqrt{1 + y^2}/|\mu| \end{aligned}$$

Therefore, Theorem 1 gives:

$$P_Y(y) = \int_{-\infty}^{\infty} |\mu| P_{X_1 X_2}(y\mu, \mu) d\mu$$

Example 3: $y = f(\mathbf{x}) = x_1^2 + x_2^2 \quad (f : \mathbf{R}^2 \rightarrow \mathbf{R})$

The set $f^{-1}(y)$ is the union of the parameterizations: $\mathbf{x}_1(\mu; y) = (\mu, \sqrt{y - \mu^2})^T$

and $\mathbf{x}_2(\mu; y) = (\mu, -\sqrt{y - \mu^2})^T$ with $\mu \in [-\sqrt{y}, \sqrt{y}]$. (The two charts overlap only at the endpoints $\mu = \pm\sqrt{y}$, a set of surface measure zero, as permitted by Theorem 1.)

$$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; y) D_\mu \mathbf{x}_1(\mu; y)|} = \|(1, -\mu/\sqrt{y - \mu^2})\| = \sqrt{y/(y - \mu^2)}.$$

$$\sqrt{|D_\mu \mathbf{x}_2^T(\mu; y) D_\mu \mathbf{x}_2(\mu; y)|} = \|(1, \mu/\sqrt{y - \mu^2})\| = \sqrt{y/(y - \mu^2)}.$$

$$\sqrt{|Df(\mathbf{x}_1(\mu; y)) Df^T(\mathbf{x}_1(\mu; y))|} = 2\|(\mu, \sqrt{y - \mu^2})\| = 2\sqrt{y}.$$

$$\sqrt{|Df(\mathbf{x}_2(\mu; y)) Df^T(\mathbf{x}_2(\mu; y))|} = 2\|(\mu, -\sqrt{y - \mu^2})\| = 2\sqrt{y}.$$

Therefore, Theorem 1 gives:

$$P_Y(y) = \begin{cases} \int_{-\sqrt{y}}^{\sqrt{y}} \frac{P_{X_1 X_2}(\mu, \sqrt{y - \mu^2}) + P_{X_1 X_2}(\mu, -\sqrt{y - \mu^2})}{2\sqrt{y - \mu^2}} d\mu & y \geq 0; \\ 0 & \text{otherwise.} \end{cases}$$

Example 4: $y_1 = x_1 + x_2 + x_3$; $y_2 = x_1 - x_2$ ($f : \mathbf{R}^3 \rightarrow \mathbf{R}^2$)

The set $f^{-1}(\mathbf{y})$ can be parameterized as $\mathbf{x}_1(\mu; y_1, y_2) = (\mu, \mu - y_2, y_1 + y_2 - 2\mu)^T$ with $\mu \in (-\infty, \infty)$.

$$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; \mathbf{y}) D_\mu \mathbf{x}_1(\mu; \mathbf{y})|} = \|(1, 1, -2)\| = \sqrt{6}$$

$$\text{And, } \sqrt{|Df(\mathbf{x}_1(\mu; \mathbf{y})) Df^T(\mathbf{x}_1(\mu; \mathbf{y}))|} = \sqrt{\left\| \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{pmatrix} \right\|} = \sqrt{6}$$

Therefore, Theorem 1 gives:

$$P_{Y_1 Y_2}(y_1, y_2) = \int_{-\infty}^{\infty} P_{X_1 X_2 X_3}(\mu, \mu - y_2, y_1 + y_2 - 2\mu) d\mu$$

Example 5a: $y_1 = x_1 + x_2 + x_3$; $y_2 = x_1 - x_2 + x_4$ ($f : \mathbf{R}^4 \rightarrow \mathbf{R}^2$)

The set $f^{-1}(\mathbf{y})$ can be parameterized as $\mathbf{x}_1(\mu_1, \mu_2; y_1, y_2) = (\mu_1, \mu_2, y_1 - \mu_1 - \mu_2, y_2 - \mu_1 + \mu_2)^T$ with $\mu_1, \mu_2 \in (-\infty, \infty)$.

$$\sqrt{|D_\mu \mathbf{x}_1^T(\mu; \mathbf{y}) D_\mu \mathbf{x}_1(\mu; \mathbf{y})|} = \sqrt{\left\| \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ -1 & 1 \end{pmatrix} \right\|} = 3$$

$$\text{And, } \sqrt{|Df(\mathbf{x}_1(\mu; \mathbf{y})) Df^T(\mathbf{x}_1(\mu; \mathbf{y}))|} = \sqrt{\left\| \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \right\|} = 3$$

Therefore, Theorem 1 gives:

$$P_{Y_1 Y_2}(y_1, y_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P_{X_1 X_2 X_3 X_4}(\mu_1, \mu_2, y_1 - \mu_1 - \mu_2, y_2 - \mu_1 + \mu_2) d\mu_1 d\mu_2$$

Example 5b: $y_1 = x_1 + x_2 + x_3$; $y_2 = x_1 - x_2 + x_4$ ($f : \mathbf{R}^4 \rightarrow \mathbf{R}^2$)

This is the same function as in Example 5a. The difference in this example is how we parameterize the set $f^{-1}(\mathbf{y})$. In this example we parameterize this set in the following way: $\mathbf{x}_1(\mu_1, \mu_2; y_1, y_2) = (2\mu_1, \mu_2, y_1 - 2\mu_1 - \mu_2, y_2 - 2\mu_1 + \mu_2)^T$ with $\mu_1, \mu_2 \in (-\infty, \infty)$.

Continuing as in 5a we have: $\sqrt{|D_\mu \mathbf{x}_1^T(\mu; \mathbf{y}) D_\mu \mathbf{x}_1(\mu; \mathbf{y})|} = \sqrt{\left| \begin{pmatrix} 2 & 0 & -2 & -2 \\ 0 & 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ -2 & -1 \\ -2 & 1 \end{pmatrix} \right|} = 6$

And, $\sqrt{|Df(\mathbf{x}_1(\mu; \mathbf{y})) Df^T(\mathbf{x}_1(\mu; \mathbf{y}))|} = \sqrt{\left| \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \right|} = 3$

Therefore, Theorem 1 gives:

$$P_{Y_1 Y_2}(y_1, y_2) = 2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P_{X_1 X_2 X_3 X_4}(2\mu_1, \mu_2, y_1 - 2\mu_1 - \mu_2, y_2 - 2\mu_1 + \mu_2) d\mu_1 d\mu_2$$

Substituting $s = 2\mu_1$ shows that this equals the result of Example 5a. The two parameterizations cover the same manifold, and the surface-measure factor $\sqrt{|D_\mu \mathbf{x}^T D_\mu \mathbf{x}|}$ exactly compensates for the change of parameterization: formula (2) is independent of the choice of charts.

Example 6: $y = \sum_{i=1}^n x_i^2 \quad (f : \mathbf{R}^n \rightarrow \mathbf{R})$

We assume that the x_i are independent normals with mean 0 and variance σ^2 . When $\sigma = 1$, y has a Chi-squared distribution of order n . We derive a generalization of this fact using our formula. The set $f^{-1}(y)$ is a sphere in $\mathbf{R}^n$ of radius $\sqrt{y}$. The density function of y is:⁴

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{P_{\mathbf{X}}(\mathbf{x}) dS}{\sqrt{|Df(\mathbf{x}) Df^T(\mathbf{x})|}}$$

Here, $Df(\mathbf{x}) = 2\mathbf{x}^T$. Consequently, on the sphere of radius $\sqrt{y}$ we have $\sqrt{|Df(\mathbf{x}) Df^T(\mathbf{x})|} = 2\sqrt{y}$. Therefore, we may write the above as

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{P_{\mathbf{X}}(\mathbf{x}) dS}{2\sqrt{y}}$$

This is

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{e^{-\sum_{i=1}^n x_i^2 / (2\sigma^2)} dS}{(2\pi)^{n/2} \sigma^n 2\sqrt{y}}$$

Which is

$$P_Y(y) = \int_{\partial B(\sqrt{y})} \frac{e^{-y/(2\sigma^2)} dS}{(2\pi)^{n/2} \sigma^n 2\sqrt{y}}$$

Or,

$$P_Y(y) = \frac{e^{-y/(2\sigma^2)}}{(2\pi)^{n/2} \sigma^n 2\sqrt{y}} \int_{\partial B(\sqrt{y})} dS$$

The surface area of a sphere of radius $\sqrt{y}$ in n dimensions is $2y^{(n-1)/2} \frac{\pi^{n/2}}{\Gamma(n/2)}$. Therefore, $P_Y(y)$, may be written:

$$P_Y(y) = \frac{y^{n/2-1} e^{-y/(2\sigma^2)}}{2^{n/2} \sigma^n \Gamma(n/2)} \quad y > 0,$$

⁴ We use the surface-measure form of the transformation formula, equation (1), with $f^{-1}(y)$ the sphere of radius $\sqrt{y}$; no global parameterization of the sphere is needed.

with $P_Y(y) = 0$ for $y \leq 0$. When $\sigma = 1$ this is the Chi-squared density with n degrees of freedom, as claimed.

Application to the Distribution of the Ratio of Normals

We use Theorem 1 to compute the distribution of the ratio of two correlated normals.

If x_1, x_2 are Gaussian random variables with joint density function, $P_{X_1 X_2}$, means: μ_1, μ_2 , and positive definite covariance matrix:⁵ $\Sigma = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, then the density function of the ratio $y = x_1/x_2$ is given by the formula in Example 2 (this density was studied classically by Geary [3] and Hinkley [4]):

$$P_Y(y) = \int_{-\infty}^{\infty} |s| P_{X_1 X_2}(sy, s) ds = \frac{1}{2\pi\sqrt{ac-b^2}} \int_{-\infty}^{\infty} |s| e^{-([sy-\mu_1, s-\mu_2]\Sigma^{-1}[sy-\mu_1, s-\mu_2]^T)/2} ds$$

Writing the exponent as $-(\alpha s^2 - 2\beta s + 2\gamma)/2$, completing the square in s , and using the identity

$$\int_{-\infty}^{\infty} |s| e^{-\alpha s^2/2 + \beta s} ds = \frac{2}{\alpha} \left[1 + \sqrt{\pi} \tau e^{\tau^2} \operatorname{erf}(\tau) \right], \quad \tau = \frac{|\beta|}{\sqrt{2\alpha}},$$

this may be reduced to:

$$P_Y(y) = \frac{\sqrt{ac-b^2} e^{-\gamma}}{\pi(cy^2 - 2by + a)} \left[1 + \sqrt{\pi} \tau(y) e^{\tau(y)^2} \operatorname{erf}(\tau(y)) \right]$$

where

$$\gamma = \frac{c\mu_1^2 - 2b\mu_1\mu_2 + a\mu_2^2}{2(ac-b^2)}$$

$$\tau(y) = \frac{|y(c\mu_1 - b\mu_2) + a\mu_2 - b\mu_1|}{\sqrt{cy^2 - 2by + a} \sqrt{2(ac-b^2)}}$$

The function $\tau(y)$ is bounded on the interval $(-\infty, \infty)$: by the positive definiteness of Σ the denominator is bounded away from 0, and as $|y| \rightarrow \infty$, $\tau(y) \rightarrow |c\mu_1 - b\mu_2|/\sqrt{2c(ac-b^2)}$. A continuous function with finite limits at $\pm\infty$ is bounded.

Note: As a consequence, this distribution has no finite moments of order ≥ 1 . This follows since $P_Y(y)$ is $O(1/y^2)$ for large $|y|$, so $\int |y|^k P_Y(y) dy$ diverges for every $k \geq 1$. *Therefore, one cannot talk about the mean or standard deviation of this distribution.*

This distribution is also not unimodal in general; Marsaglia [5] gives explicit bimodal examples.

Special Cases

The density function simplifies under the condition: $\mu_1 = \mu_2 = 0$. $P_Y(y)$ is given by:

$$P_Y(y) = \frac{\sqrt{ac-b^2}}{\pi(cy^2 - 2by + a)} = \frac{\sqrt{ac-b^2}}{\pi\left((\sqrt{c}y - \frac{b}{\sqrt{c}})^2 + (a - \frac{b^2}{c})\right)}$$

⁵ If Σ is positive definite we have: $a > 0$, $c > 0$, $ac - b^2 > 0$.

If, in addition, x_1 and x_2 are uncorrelated, then $P_Y(y)$ becomes:

$$P_Y(y) = \frac{\sqrt{ac}}{\pi(cy^2+a)}$$

Finally, if we further set, $a = c = 1, b = 0$; that is, x_1 and x_2 are independent Gaussians each with mean 0 and variance 1, then $P_Y(y)$ becomes:

$$P_Y(y) = \frac{1}{\pi(y^2+1)}$$

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